Public Policies for Innovation and Firms

Guiding Questions for the Oral Exam

Gianluca Monturano, PhD
University of Salento

The following questions are intended to guide Erasmus students in preparing for the oral exam. The exam will focus on the main economic mechanisms, policy instruments and evaluation concepts discussed during the course.

Lecture 1 – Introduction

1. Why may public intervention be needed in the field of innovation and firms?
2. What are the main types of public policies discussed in the course?

Lecture 2 – Public Economics of Innovation

1. Why can knowledge generate spillovers and appropriability problems?
2. What is the difference between the private return and the social return to innovation?
3. Why may firms invest less in innovation than would be socially desirable?

Lecture 3 – Financing Innovation

1. Why do innovative firms and startups often face financial constraints?
2. What are adverse selection and moral hazard in credit markets?
3. Why may equity or venture capital be more suitable than bank debt for some innovative firms?

Lecture 4 – Public Incentives

1. Why do governments use subsidies and tax credits to support investment and innovation?
2. What do additionality and deadweight mean?
3. What is the difference between crowding-in and crowding-out?

Lecture 5 – Public Guarantees and Venture Capital

1. How does a public credit guarantee work?
2. What are the main benefits and risks of public guarantees?
3. What is the rationale for public venture capital and public–private co-investment?

Lecture 6 – Industrial and Innovation Policy

1. Which market failures may justify industrial policy?
2. What is the difference between horizontal and vertical policies?
3. What are mission-oriented policies?
4. What are the main risks of industrial policy, such as capture, rent seeking and lock-in?

Lecture 7 – Infrastructure, Digitalisation and Territory

1. Why can infrastructure be considered a collective productive input?
2. What are network effects and why do they matter for digital technologies?
3. What is the digital divide?
4. What are agglomeration economies and why can policy effects differ across territories?

Lecture 8 – Policy Evaluation and the Counterfactual

1. What does it mean to estimate the causal effect of a public policy?
2. What is a counterfactual?
3. What are potential outcomes?
4. What is the difference between ATE and ATT?
5. Why can a simple comparison between treated and untreated units be affected by selection bias?

Lecture 9 – Difference-in-Differences

1. What is the basic intuition behind the difference-in-differences method?
2. What is the parallel trends assumption?
3. Why do we compare both treated and control units before and after the policy?
4. What is an event study and what information can it provide?

Lecture 10 – Spatial and Territorial Policy Evaluation

1. Why can spatial spillovers make policy evaluation more difficult?
2. What does contamination of the control group mean?
3. What is displacement?
4. Why can the same policy generate different effects across territories?
5. Why is the geographical scale of analysis important in policy evaluation?

General indication. Students should be able to explain the main economic mechanism, the rationale for public intervention, the advantages and limitations of the policy instrument, and the basic logic used to evaluate its effects.